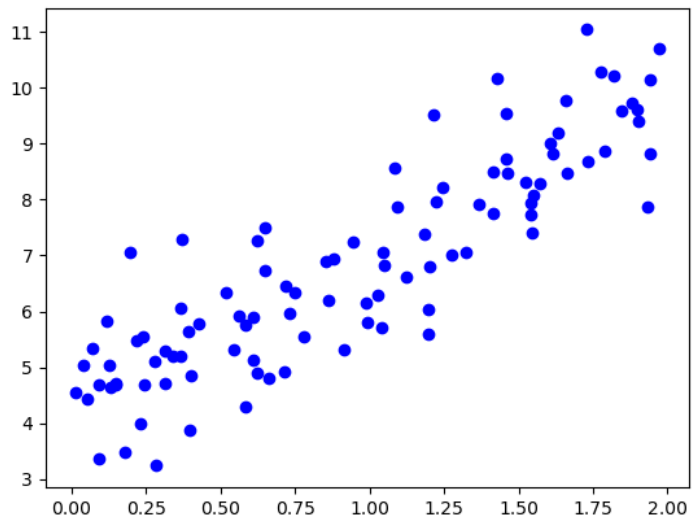
Regression and Classification (10 Scores)

Homework 2 of Deep Learning and Financial Data Analysis
Spring 2026
Deadline: 2026-03-18 22:00 (NOT 23:59!!!)

1 Linear Regression (40%)

In task 1, we provide a simple sample set of 100 points shown in figure below.



Your job is to finish the **linear_regression()** function in the given lr.py to return the slope and intercept of the equation that minimizes mean square error over the 100 points. You should do it in two ways:

(1) Directly calculating the optimal weights.

(2) Using gradient descent (or SGD) to learn the optimal weights.
You should keep code of both approaches within your submitted script. Check to see if results from the two approaches match. Your should save the plot with your fitted line and put the calculated error in your report.

2 Softmax Classification (60%)

In task 2, you are required to use **Softmax Classifier** to perform MNIST classification.
MNIST digits dataset is a widely used dataset for image classification in machine learning field. It contains 60,000 training examples and 10,000 testing examples. The digits have been size-normalized and centered in a fixed-size image. Each example is a 784×1 matrix, which is transformed from an original 28×28 grayscale image. Digits in MNIST range from 0 to 9. Some examples are shown below. **Note**: During training, information about testing examples should never be used in any form.

Note: We will use Jupyter Notebook in homework-2, please install it.

There are several files included in the `./homework2/` folder:

- `lr.py` describes the linear regression. You are required to complete the function `linear_regression(x, y)`. You only need to write down your code after **# TODO** to calculate the slope and intercept of your linear regression equation.
- `data.npy` is the file contains 100 sample points. The loading function is provided, no need to check it.
- `homework2-softmax.ipynb` is an IPython Notebook file which describes the main contents of this homework. Data loading, hyperparameters setting, training and testing are included in this file. **Please read this file carefully**.
- `mnist_data_loader.py` is used to load MNIST dataset. Not required reading.
- `softmax_classifier.py` describes the softmax classifier. **You are required to complete** the function `softmax_classifier(W, input, label, lamda)`. Part of this function is provided and you need to write down your code after **# TODO** to calculate the loss, gradientand prediction.

3 Requirements

You are required to complete the **# TODO** parts in above files. **You need to submit all codes and a short report** with the following requirements:

- Record the mean square error of your linear regression equation.
- Include the plot of your linear regression equation with 100 points in your report.
- Record the training and testing accuracy, plot the training loss curve and training accuracy curve in the report.
- The given hyperparameters may not perform very well. Modify them by your own, and observe how do they affect the classification performance. Write down the results and your observation in the report.
- Make sure your code is runnable. A code would be graded as 0 if it has any running error.

4 Attention

- You need to submit **all codes** and **a short report** (at least one page **in PDF format**). Delete data.npy and the MNIST dataset before submit.
- Do not paste a lot of codes in your report (only some essential lines should be included).
- Pay attention to the efficiency of your implementation. Try to finish this homework without the use of **for-loops**, using matrix multiplication instead.
- Any open source neural network toolkits, such as TensorFlow, Caffe, PyTorch, are **NOT** permitted in finishing homework 2 except for data preprocessing.
- **Plagiarism is not permitted**.